Ex - 04	EDA-Data Cleaning

AIM

To perform data cleaning using Python Pandas by detecting, filling, and dropping missing values, removing duplicate records and unnecessary data, converting data types, ensuring data consistency, and normalizing numerical data using standardization and Min-Max scaling.

PROBLEM STATEMENT

To clean and preprocess a dataset using Python by handling missing values, eliminating duplicate records, removing unnecessary columns, converting data into appropriate data types, and applying normalization techniques to improve data quality and prepare the dataset for further analysis.

PROCEDURE

1. Import the required libraries, including Pandas and Scikit-learn.
2. Create a sample dataset containing missing values, duplicate records, and inconsistent data types.
3. Display the original dataset and identify missing values using isnull().sum().
4. Fill missing numerical values using the mean and categorical values using the mode.
5. Drop rows containing missing values where appropriate.
6. Remove duplicate records using drop_duplicates().
7. Remove unnecessary columns that are not required for analysis.
8. Convert columns into appropriate data types and standardize inconsistent text values.
9. Apply StandardScaler and MinMaxScaler to normalize numerical features.
10. Display the cleaned dataset and verify that missing values and duplicates have been handled.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Step 1: Import dataset
df = pd.read_csv('Student_Perfomance.csv')

print("Original Dataset:")
print(df.head())

# Step 2: Select required columns
columns = ['math score', 'reading score', 'writing score']

data = df[columns].copy()

# Step 3: Handle missing values
data = data.fillna(data.mean())

# Use first 20 students for visualization
data = data.head(20).reset_index(drop=True)

# Step 4: Min-Max Normalization
minmax_scaler = MinMaxScaler()
minmax_data = pd.DataFrame(
    minmax_scaler.fit_transform(data),
    columns=columns
)

# Step 5: Standardization
standard_scaler = StandardScaler()
standard_data = pd.DataFrame(
    standard_scaler.fit_transform(data),
    columns=columns
)

# Step 6: Function to create separate bar + line plots
def plot_data(dataset, method):
    for column in columns:
        plt.figure(figsize=(8, 5))

        x = range(1, len(dataset) + 1)
        values = dataset[column]

        # Bar plot
        plt.bar(x, values, alpha=0.6, label='Bar Plot')

        # Line plot
        plt.plot(
            x, values,
            color='red',
            marker='o',
            linewidth=2,
            label='Line Plot'
        )

        plt.title(f'{method} - {column.title()}')
        plt.xlabel('Student Number')
        plt.ylabel(column.title())
        plt.legend()
        plt.grid(axis='y', linestyle='--', alpha=0.4)
        plt.tight_layout()
        plt.show()

# Step 7: Original data plots (3)
plot_data(data, 'Original Data')

# Step 8: Min-Max normalization plots (3)
plot_data(minmax_data, 'Min-Max Normalization')

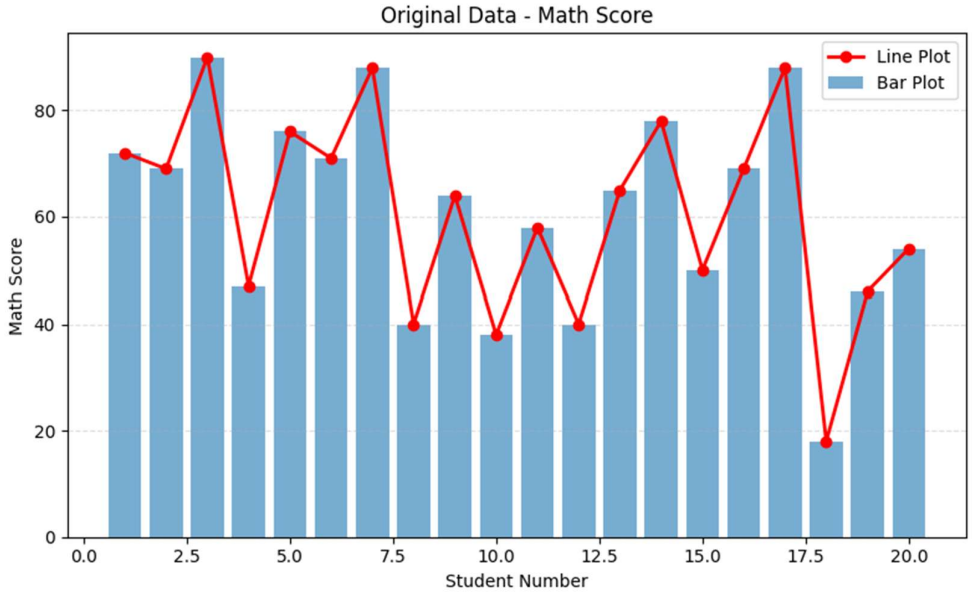
# Step 9: Standardization plots (3)
plot_data(standard_data, 'Standardization')

# Step 10: Save output files
data.to_csv('Original_Data.csv', index=False)
minmax_data.to_csv('MinMax_Normalized.csv', index=False)
standard_data.to_csv('Standardized_Data.csv', index=False)

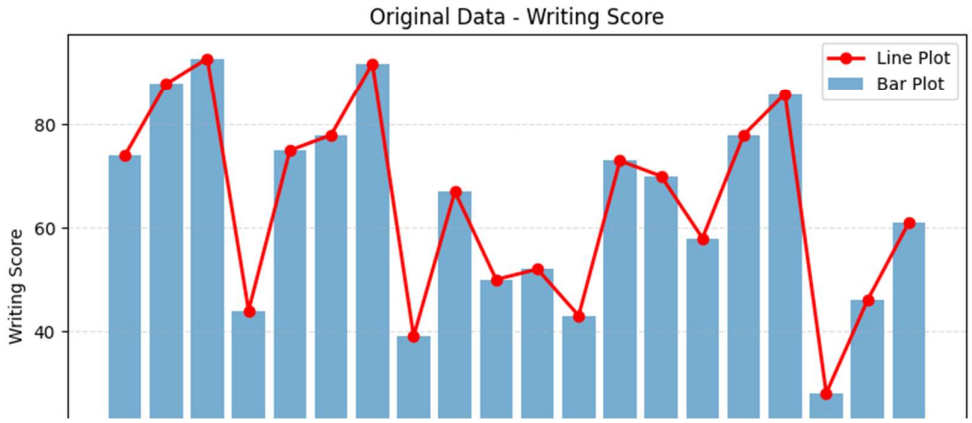
print("All 9 plots generated successfully!")
print("Output CSV files saved successfully!")
```

```
Original Dataset:
   gender race/ethnicity parental level of education         lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B             master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test preparation course  math score  reading score  writing score
0                    none          72             72             74
1               completed          69             90             88
2                    none          90             95             93
3                    none          47             57             44
4                    none          76             78             75
```


Original Data - Math Score
Line Plot
Bar Plot
Math Score
Student Number
0
20
40
60
80
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Original Data - Reading Score
Line Plot
Bar Plot
Reading Score
Student Number
0
20
40
60
80
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Original Data - Writing Score
Line Plot
Bar Plot
Writing Score
40
60
80

Expected output (summary)

- Missing values in Age and Marks are filled using their mean values.
- Duplicate records are removed.
- The unnecessary column is deleted.
- Text values are made consistent, and data types are converted.
- Standardized marks have a mean approximately equal to 0 and standard deviation approximately equal to 1.
- Min-Max normalized marks fall between 0 and 1.

RESULT

Data cleaning was successfully performed using Python Pandas and Scikit-learn. Missing values were handled, duplicate records and unnecessary columns were removed, data types and text values were standardized, and numerical data was normalized using StandardScaler and MinMaxScaler. The resulting dataset is cleaner and suitable for further exploratory data analysis and machine learning.